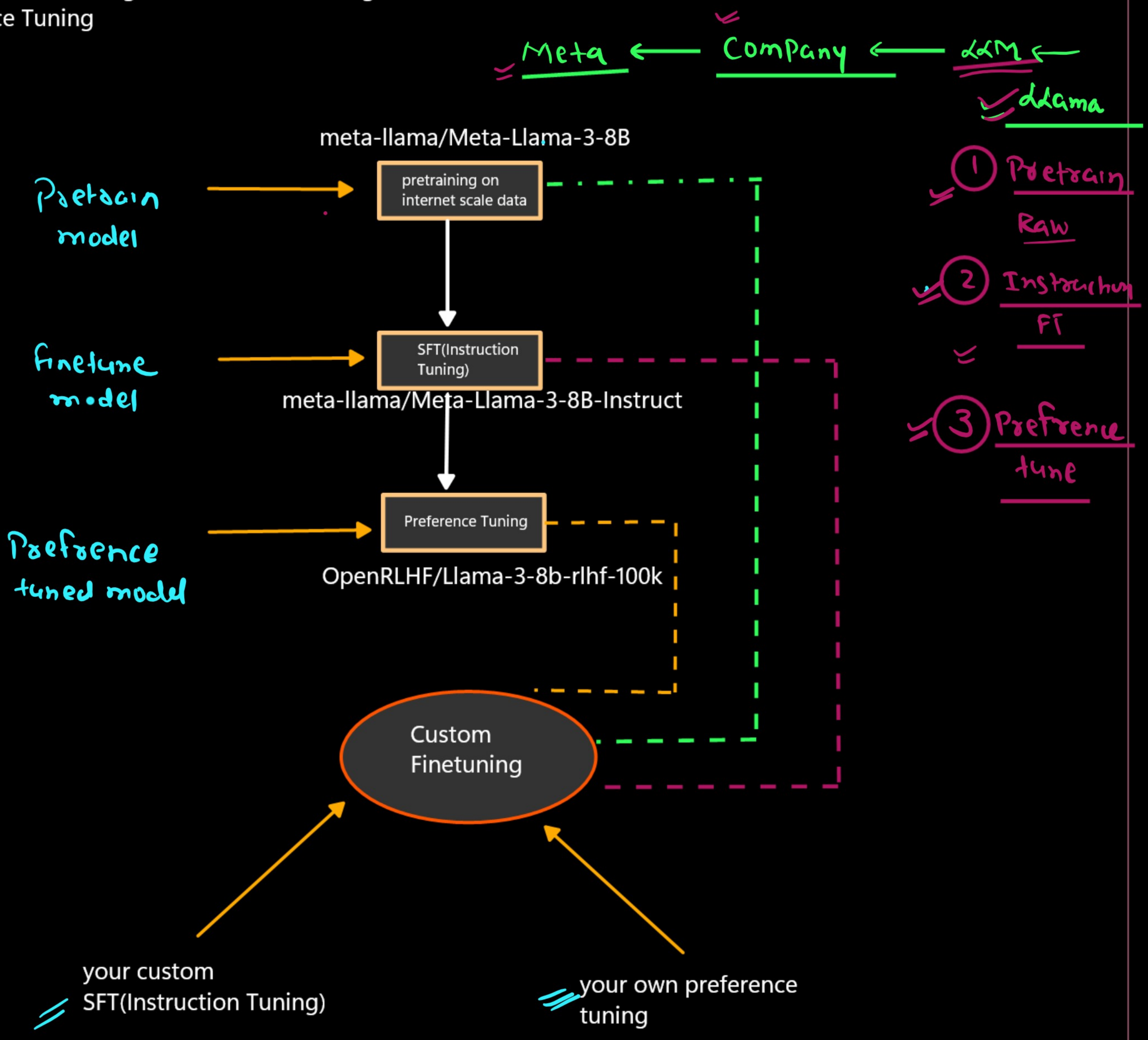
Fine-tuning

Fine-tuning is the process of taking a already trained model and training it further on domain-specific or task-specific data so that it performs better for a particular use case.

LLM Training Pipeline
1. Pretraining of model
2. Supervised Finetuning(Instruction Finetuning)
3. Preference Tuning



1. hugging face Modules
2. unsloth

1. FULL Finetuning
2. Partial Finetuning(PEFT)

LLM → 1. Meta
2. Mistral

{ train the model on your own data }

customtuning

1. Pretraining ← RAW
2. SFT | Instruction ← I/P ⟷ O/P
3. Preference tuning

meta-llama/Meta-Llama-3-8B

↓

meta-llama/Meta-Llama-3-8B-Instruct
(how to follow the instruction)

usecase: Pharama --> document on molecule study
1. to train a llm from scratch it is difficult
2. i will take any opensource model i will download it on my owm server
3. and then i will finetune that

my requirement was just to generate the data not the conversation AI/Chatbot/QA

my requirement was just to generate the data
meta-llama/Meta-Llama-3-8B --> i have trained(finetuning) on my own pharama datset

conversation AI/Chatbot/QA
meta-llama/Meta-Llama-3-8B-Instruct --> further i will finetuning it on my own dataset
IP--OP

model → Parameter (weight | bias) (trained)

transformer → Self Attention ← Q | K | V ← weight
FFNN ← weight

1. Pretraining (Unsupervised | Self supervised learning)

2. SFT / IFT → Supervised finetuning (I/P ⟷ O/P)

Parameter level

1. ~~Full finetuning~~ ← Retrain all the Parameter (weight & bias)
2. Partial finetuning ← Subset of Parameter

1. Huge GPU Power 2. huge infra

Two method

1. OLD School method

CV
- CNN → VGG | Resnet ...
- BERT
- BART
- T5

retrain

1. Freeze all the layer train only last O/P layer
2. Freeze some starting layer & retrain the remaining last layer

(Latest ← LLM ← huge architecture)

Latest

2. PEFT ← Small Infra

Single GPU | Small VRAM

Parameter efficient FT

1. LoRA → QLoRA — Low Rank Adaption

ADAPTER

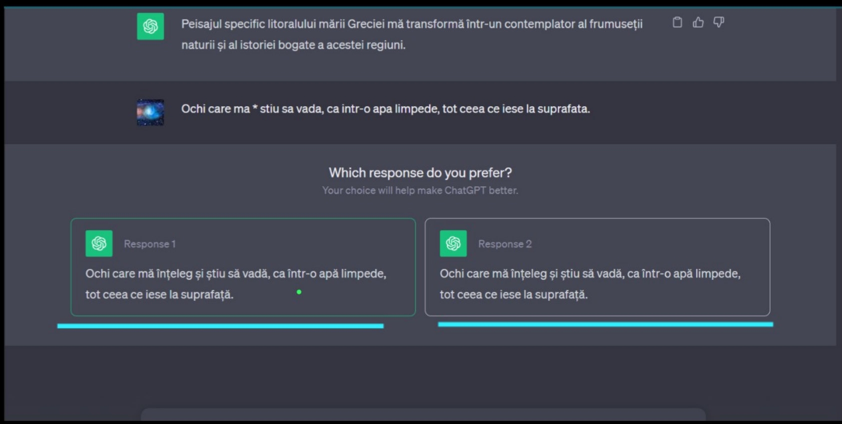
Quantized (Float → small float → int)

2. DoRA — (weight - decomposition)

3. BitFit

4. IA^3

3 Prefrence alignment



collect

Again we train

Retrain our model

1 $\boxed{\text{RLHF}}$ → OpenAI → RL → DRL → $\boxed{\text{PPO}}$

framework

2 GRPO

3 $\boxed{\text{DPO}}$ (~~RL~~) → 4 ORPO

Hugging Face is an AI company that provides a complete ecosystem for AI development

Mainly including training LLMs from scratch, fine-tuning, evaluation, hosting applications.

almost the entire modern LLM fine-tuning industry workflow revolves around the Hugging Face ecosystem.

In modern Hugging Face ecosystem: Most Important Libraries
“If your goal is only modern LLM fine-tuning...
Then these libraries are MUST KNOW:
Transformers
Datasets
Tokenizers
PEFT
Accelerate
TRL
Bitsandbytes
Safetensors
Evaluate
Lighteval

Evaluation Matrix:
accuracy
F1-score
BLEU
ROUGE
perplexity
benchmark scores

Native vs Outsider Library
“Now here is an interesting point...
Not every tool is originally built by Hugging Face.”

Native HF Libraries

Transformers
Datasets
Tokenizers
PEFT
Accelerate
TRL
Diffusers
Evaluate
Optimum
Safetensors

Outsider but Integrated
Bitsandbytes
timm
Sentence Transformers
Argilla
Distilabel

So the ecosystem is integrated, even if ownership is different.

→ Datasets Library/ Custom Dataset
→ Preprocessing / Cleaning
→ Tokenizer (Tokenized Dataset)
→ Transformers Model Loading
→ PEFT (LoRA Adapters)
→ Bitsandbytes (4-bit / Quantization)
→ Accelerate (Distributed & Multi-GPU Training)
→ Trainer / TRL (SFT, RLHF, DPO, PPO, GRPO)
→ Evaluation (Accuracy, BLEU, ROUGE, Perplexity, Benchmarks)
→ Save Checkpoints (Safetensors)
→ Push to Hugging Face Hub
→ Inference / Deployment (TGI, TEI, Inference Endpoints)
→ Monitoring & Iteration

Accuracy / Quality Monitoring
Latency Monitoring
GPU / CPU Resource Monitoring
Token Usage / Cost Monitoring

Add new training data
Better LoRA adapters
RLHFDPO tuning
Hyperparameter tuning

Hugging Face is the foundation of the Finetuning ecosystem.
And Unsloth is an ultra-optimized performance layer built on top of that foundation which enables:
faster training
lower VRAM usage
longer context training
cheaper fine-tuning

Now Enters Unsloth

"Everything we discussed so far...
was the standard Hugging Face workflow.

But there was some huge issue::
slow
consumes huge VRAM
expensive

And that is exactly the problem solved by:
UNSLOTH."

Unsloth Definition

"Unsloth is an ultra-optimized LLM fine-tuning framework built on top of the Hugging Face ecosystem that enables fast LoRA/QLoRA/RLHF training with very low VRAM usage."

Main Claims of Unsloth

"Unsloth claims:

2x to 3x faster training
50–80% less VRAM usage
Ability to train large models even on free GPUs"
Real Practical Meaning

"For example:

If a Hugging Face workflow takes:

10 hours
40 GB VRAM

Then Unsloth may complete the same task in:

~5 hours
12–16 GB VRAM."
LLaMA Example

"Take the example of LLaMA 3.1 8B:

Standard HF:
24 GB VRAM

Unsloth:
only ~7–8 GB."

“The most shocking feature of Unsloth is:

LONG CONTEXT TRAINING.”

20K–300K+ token training becomes possible.

“Where standard HF gives OOM…
Unsloth can handle extremely long contexts.”

GPU vs Context Length

“Example:
LLaMA 3.1 (8B)

12 GB → ~21K tokens
16 GB → ~40K
24 GB → ~78K tokens
80 GB → ~340K tokens

Whereas standard HF:
Even on an 80 GB GPU, only ~28K tokens are supported.”

But HOW is Unsloth So Fast?

"Now the question is…

How is this magic happening?"

Answer:
Deep low-level GPU optimizations.

CUDA and Triton

"CUDA is NVIDIA's low-level GPU programming layer.

It:

runs on GPUs
is massively parallel
is extremely fast
provides low-level control

And Triton is a Python-like GPU language that generates highly optimized CUDA kernels."

Internal Optimizations

"Internally, Unsloth uses:

custom CUDA kernels
Triton kernels
fused attention
optimized forward/backward pass
smart checkpointing
Flash Attention
manual backpropagation
automatic sequence packing"

“The most important point:

Unsloth does not rely on approximation tricks.

It maintains exact math.

That is why:

accuracy loss is negligible
model quality does not degrade”
Architecture Flow

$2.3456 \times 8.9123 = 20.90518688$
$2.3 \times 8.9 \approx 20.47$

“Exact math means Unsloth improves speed and memory efficiency without changing the actual mathematical correctness of the model computations.”

“Internally, the stack looks something like this:”

Unsloth
↓
Transformers + PEFT + TRL
↓
PyTorch
↓
CUDA / Triton
↓
GPU

“This means Unsloth does not replace Hugging Face.

Instead, it works like an optimized engine on top of Hugging Face.”